FOOT & ANKLE · NEET-SS / FET / INI-SS

Talus and calcaneus

Part A: Talus fractures

1. Anatomy

- **About 60% of the talus is covered by articular cartilage**; **no muscle or tendon attachments**; regions: **head, neck, body (dome), lateral process, posterior process (medial and lateral tubercles; FHL groove between them)**.
- **Blood supply (retrograde to the body, like the scaphoid):**
  - **Artery of the tarsal canal (from the posterior tibial artery): the main supply to the talar body (about 50%)**,
  - **Deltoid branch** (from the artery of the tarsal canal: **supplies the medial body; may be the only remaining supply after neck fractures: preserve the deltoid ligament**),
  - **Artery of the tarsal sinus (from the peroneal and dorsalis pedis arteries)**: lateral body,
  - **Dorsalis pedis branches**: head and neck (superior neck).
  - **Anastomosis of the tarsal canal and sinus arteries beneath the neck (tarsal sling)**.
- **Neck fractures disrupt intraosseous supply; dislocation of the subtalar, ankle and talonavicular joints sequentially destroys the remaining vessels**.

2. Talar neck fractures

- **Mechanism: forced dorsiflexion (axial load): "aviator's astragalus"** (neck impacts the anterior tibia); **high-energy (RTA, falls)**; **about 50% of talus fractures**.

- **Hawkins classification (modified by Canale and Kelly; Vallier subdivision of II):**

Type	Description	AVN risk (approximate)
I	Undisplaced neck fracture	0-15%
II	Displaced + subtalar subluxation/dislocation (IIA subtalar subluxed; IIB subtalar dislocated (Vallier: IIB higher AVN))	20-50%
III	Displaced + subtalar AND ankle (tibiotalar) dislocation (body extruded, usually posteromedially)	50-100% (about 70-90%)
IV	III + talonavicular subluxation/dislocation	Up to about 100%

- **Imaging:** radiographs (**Canale view: 15-degree pronated, 75-degree cephalad-tilted AP of the foot: best view of the talar neck**), **CT (essential: comminution, displacement)**.
- **Management:**
  - **Type I (truly undisplaced, confirmed on CT): non-weight-bearing cast 6-12 weeks** (or fixation to allow early motion).
  - **Displaced: urgent reduction of dislocations (emergency: skin at risk, especially with extruded body)**; **definitive ORIF can be delayed until soft tissues allow** (**timing of fixation does not influence AVN rates (Vallier, Lindvall); the reduction of dislocations is the emergency**).
  - **ORIF via dual anteromedial + anterolateral approaches** (**accurate assessment of reduction: avoid varus malreduction (medial comminution)**), **screws (posterior-to-anterior screws biomechanically stronger) ± mini-fragment plates (lateral/medial bridge plates for comminution)**; **medial**

malleolar osteotomy for body access (**preserve the deltoid ligament: deltoid branch**).

- **Complications:**
  - **AVN** (**Hawkins sign: subchondral lucency (radiolucent band) in the talar dome on the AP ankle at 6-8 weeks = revascularisation (bone resorption implies blood supply): good prognostic sign; absence does not guarantee AVN**), **MRI**; **AVN does not always progress to collapse**; **creeping substitution**; **protected weight bearing debated**.
  - **Post-traumatic arthritis (subtalar commonest: about 50-100%)**, **ankle arthritis**,
  - **Malunion (varus malunion commonest: medial comminution; leads to hindfoot varus, lateral column overload, reduced subtalar eversion)**,
  - **Non-union (about 5-10%)**, **infection, skin necrosis (open fractures: about 20-25% of neck fractures are open)**.
  - **Salvage: subtalar fusion, TTC/Blair fusion (tibiotalar fusion with anterior sliding tibial graft into the talar neck/head: for AVN of the body), talectomy (historical)**.

3. Talar body, process and head fractures

- **Talar body fractures** (**intra-articular; ankle and subtalar**): **ORIF often needs malleolar osteotomy**; **AVN and arthritis common**.
- **Lateral process fractures ("snowboarder's fracture")**: **dorsiflexion + inversion/eversion with axial load**; **missed as ankle sprains (pain just anterior and inferior to the tip of the lateral malleolus)**: **CT**; **undisplaced/small: cast**; **large (over 1 cm) displaced: ORIF**; **comminuted: excision**.
- **Posterior process fractures:**
  - **Lateral tubercle (Shepherd fracture)**: **forced plantarflexion**; differentiate from **os trigonum**,

  - **Medial tubercle (Cedell fracture)**: **pronation-dorsiflexion (avulsion of the posterior tibiotalar deltoid)**.
  - **Non-weight-bearing cast; excision of symptomatic non-union fragments.**
- **Talar head fractures**: rare; **compression injuries**; **ORIF or excision of small fragments**.
- **Osteochondral lesions** (see ankle notes).

4. Subtalar and total talar dislocations

- **Subtalar (peritalar) dislocation: simultaneous dislocation of the talocalcaneal and talonavicular joints** with an intact tibiotalar joint.
  - **Medial (about 80%): "acquired clubfoot" appearance (inversion)**; **"basketball foot"**; **irreducible by: talar head buttonholed through the extensor digitorum brevis, extensor retinaculum or talonavicular capsule, peroneal tendons**.
  - **Lateral (about 20%): higher energy, worse prognosis; "acquired flatfoot"**; **irreducible by: tibialis posterior tendon (interposed: commonest block), FHL, FDL**.
  - **Urgent closed reduction (knee flexed to relax the gastrocnemius, accentuate deformity, then reverse)**, **open reduction if irreducible**; **CT after reduction** (associated fractures in about 50%+); **cast about 4-6 weeks**; **subtalar stiffness common**.
- **Total talar extrusion (dislocation)**: **open injury usually**: **reimplant the talus after thorough debridement** (lower complication rates than talectomy); **AVN almost certain** but reimplantation preserves height and function.

Part B: Calcaneus fractures

5. Anatomy and mechanism

- **Commonest tarsal bone fractured (about 60% of tarsal fractures)**; **intra-articular in about 75%**.
- **Articular facets: posterior (largest, main weight-bearing)**, **middle (on the sustentaculum tali)**, **anterior**; **sustentaculum tali (constant fragment: attached to the talus by the interosseous and deltoid ligaments; FHL runs beneath it)**; **anterior process** (calcaneocuboid).
- **Mechanism: axial load (fall from height, RTA)**: the **lateral process of the talus acts as a wedge**, creating a **primary fracture line** (anterolateral to posteromedial: **separating the sustentacular (superomedial) fragment from the tuberosity (superolateral)**) and **secondary fracture lines** (**tongue-type vs joint-depression type**: Essex-Lopresti).
- **Associated injuries (about 10% lumbar spine fractures (L1), bilateral calcaneal fractures in about 5-10%, other lower limb fractures)**: **"lover's fracture"/"don Juan" fracture pattern**; **compartment syndrome of the foot (about 10%: calcaneal compartment)**: **claw toe deformity late**.

6. Imaging and measurements

- **Lateral radiograph:**
  - **Böhler angle** (**normal 20-40 degrees**; **reduced/negative = depressed posterior facet: severity indicator, prognostic**),
  - **Critical angle of Gissane** (**normal 95-105 degrees (100-130 in some texts)**; **increased** with posterior facet depression),
  - **Double density sign** (depressed posterior facet fragment).
- **Axial (Harris) view: varus, widening, shortening of the tuberosity**; **AP foot**: CC joint involvement; **Broden views**: posterior

facet.

- **CT (essential): Sanders classification (coronal images through the widest part of the posterior facet)**:

Type	Description	Prognosis
I	**Undisplaced (under 2 mm)** posterior facet fracture (any number of lines)	**Non-operative**
II	**Two-part** fracture of the posterior facet (**IIA lateral, IIB central, IIC medial** fracture line)	**Best results with ORIF**
III	**Three-part** fracture with a **depressed central fragment** (IIIAB, IIIAC, IIIBC)	Intermediate
IV	**Four or more parts: highly comminuted**	**Poor: primary subtalar arthrodesis considered**

- **Essex-Lopresti classification (lateral radiograph): joint-depression type** (posterior facet separate from the tuberosity) vs **tongue-type** (**secondary fracture line exits posteriorly through the tuberosity: the posterior facet is attached to the tongue fragment**: **posterior skin at risk: urgent reduction/percutaneous fixation (Essex-Lopresti manoeuvre)**).

7. Management

- **Soft tissues: the key determinant of timing**: **wait for the "wrinkle sign" (return of skin wrinkles: about 7-14 days)**; **fracture blisters (blood-filled = deeper injury: avoid incisions through them)**; elevation, foot pump.
- **Emergencies: open fractures** (often medial wound), **compartment syndrome**, **tongue-type fractures with posterior skin tenting** (**reduce within 24 hours**), **fracture-dislocations**.
- **Non-operative: Sanders I**, **extra-articular fractures (minimal displacement)**, **elderly, smokers, diabetes, PVD, poor compliance,**

severe soft-tissue injury, non-ambulators: **early ROM, non-weight bearing 6-12 weeks**.

- **Operative (displaced intra-articular, Sanders II-III; tuberosity displacement):**
  - **Extensile lateral L-shaped approach** (**full-thickness flap; lateral calcaneal artery supplies the corner: wound complications/dehiscence in about 10-25%**; **sural nerve**): **ORIF with lateral plate**: **historical standard**,
  - **Sinus tarsi (minimally invasive) approach** (**fewer wound complications (about 2-5%)**, similar reduction for Sanders II-III) **(now preferred)**,
  - **Percutaneous reduction and screw fixation** (tongue-type, Sanders II),
  - **Primary subtalar arthrodesis (with ORIF of the body shape)** for **Sanders IV**.
- **Goals: restore the posterior facet congruity, Böhler angle (height), calcaneal width, length, hindfoot alignment (correct varus)**.
- **Evidence:**
  - **Buckley (Canadian RCT 2002): no overall difference between operative and non-operative treatment**; **surgery benefited: women, younger patients, non-workers' compensation, lighter work, Böhler angle 0-14 degrees, Sanders II**; **non-operative patients 5.5x more likely to need late subtalar fusion**.
  - **UK HeFT trial (Griffin 2014): no benefit of ORIF (extensile lateral) at 2 years; more complications in the surgical group**.
  - **Smokers, diabetics, workers' compensation: worse outcomes and higher wound complications (smoking: about 3x wound complications)**.

8. Complications

- **Wound dehiscence/infection** (**extensile lateral approach; smokers, diabetes, open fractures, delayed surgery under 7 days with swelling**), **sural nerve injury**.
- **Subtalar arthritis (commonest late problem)**: **subtalar arthrodesis (in situ)**; with **loss of height (horizontal talus, anterior ankle impingement, loss of dorsiflexion)**: **subtalar bone-block distraction arthrodesis (restores talar declination)**.
- **Lateral wall blowout**: **calcaneofibular (subfibular) impingement**, **peroneal tendon impingement/dislocation (about 30%)** and **stenosis**: **lateral wall exostectomy, peroneal tendon decompression**.
- **Malunion: widening (heel widening: shoe problems), shortening, varus hindfoot**: **calcaneal osteotomy + subtalar fusion** (Stephens-Sanders classification of malunion: **I lateral wall exostosis, II + subtalar arthritis, III + varus malalignment**).
- **Compartment syndrome** (missed: **claw toes, intrinsic contracture**), **heel pad pain (fat pad damage)**, **CRPS**, **tibial nerve entrapment (tarsal tunnel)**.

9. Extra-articular calcaneal fractures

- **Anterior process fractures** (**avulsion of the bifurcate ligament with inversion/plantarflexion: mimics sprain**; **small: boot/cast; large (over 25% CC joint) displaced: ORIF; symptomatic non-union: excision**).
- **Calcaneal tuberosity avulsion fractures** (**Achilles avulsion; diabetics, osteoporosis: "beak fracture"**): **posterior skin at risk: emergency reduction and fixation**.
- **Sustentaculum tali fractures** (isolated; **FHL entrapment**): ORIF if displaced.

- **Body fractures not involving the subtalar joint**: usually non-operative unless significant deformity.
- **Calcaneal stress fractures** (**military recruits, runners: sclerotic line parallel to the posterior border**): rest.
- **Paediatric calcaneus fractures**: **more extra-articular, remodel**: mostly non-operative.

Part C: Midfoot injuries (brief link)

- **Lisfranc (tarsometatarsal) injury: Lisfranc ligament: from the medial cuneiform to the base of the second metatarsal (plantar band strongest)**; **plantar ecchymosis sign**, **"fleck sign" (avulsion at the base of the second metatarsal)**; **weight-bearing radiographs: diastasis between the first and second metatarsal bases (over 2 mm)**, **medial borders of the second metatarsal and middle cuneiform should align (AP)**, **fourth metatarsal and cuboid (oblique)**; **Myerson classification (A total incongruity, B partial (B1 medial, B2 lateral), C divergent)**; **anatomic reduction essential**: **ORIF (bridge plating/screws) vs primary arthrodesis (medial 2-3 TMT joints) for purely ligamentous injuries (Ly and Coetzee: primary fusion gave better outcomes and fewer reoperations)**; **lateral column (4th/5th TMT) never fused (mobility essential): K-wire temporarily**.
- **Navicular fractures** (**body: Sangeorzan types; stress fractures: central third**), **cuboid "nutcracker" fractures** (forced abduction: **restore lateral column length**).
- **Fifth metatarsal base fractures: zone 1 (tuberosity avulsion: peroneus brevis/lateral band of plantar fascia: symptomatic treatment)**, **zone 2 (Jones fracture: metaphyseal-diaphyseal junction (4th-5th intermetatarsal joint): watershed: non-weight-bearing cast 6-8 weeks or intramedullary screw in athletes)**, **zone 3 (proximal diaphyseal stress fracture: screw fixation often)**.

Exam pearl: **Talus: artery of the tarsal canal (posterior tibial) = main body supply; deltoid branch: preserve the deltoid. Hawkins I-IV AVN rising to near 100%; Hawkins sign (subchondral lucency at 6-8 weeks) = revascularisation. Canale view for the neck. Reduce dislocations urgently; ORIF timing doesn't change AVN. Varus malunion from medial comminution. Subtalar arthritis commonest.**

Exam pearl: **Lateral process = snowboarder's fracture (mimics sprain). Medial subtalar dislocation blocked by EDB/talar head buttonhole; lateral blocked by tibialis posterior.**

Exam pearl: **Calcaneus: Böhler 20-40 (reduced), Gissane 95-105 (increased). Sanders CT (coronal, posterior facet): I non-op, II best ORIF results, IV consider primary subtalar fusion. Tongue-type with skin tenting = urgent. Wait for wrinkle sign; sinus tarsi approach fewer wound problems. Buckley: no overall difference (women, young, non-compensation benefit); HeFT: no benefit. Late: subtalar arthritis, lateral wall impingement/peroneals; loss of height = bone-block distraction fusion. Check L-spine (10%).**

Exam pearl: **Lisfranc: plantar ecchymosis, fleck sign, 1-2 MT base diastasis; ligamentous = primary medial column fusion; never fuse 4th/5th TMT. Fifth MT: zone 1 avulsion (symptomatic), zone 2 Jones (watershed: cast NWB or IM screw).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.